Descriptive Statistics

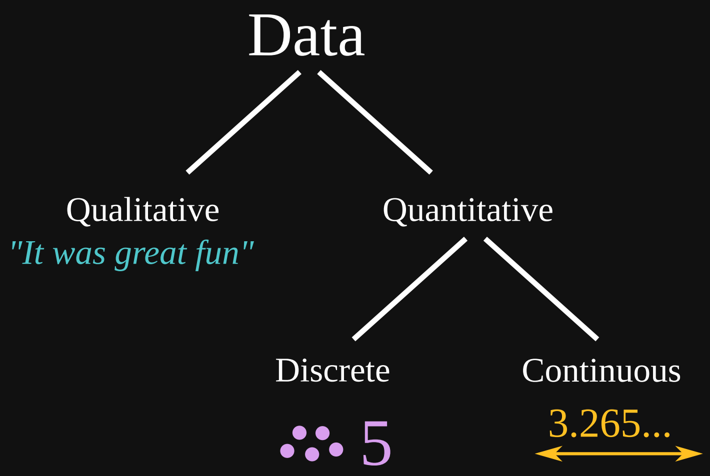
Visualization

DATA

terms

- **data**: collection of facts
- **datum**: is sigular form of data

data types



numbers, words, measurements, observations, descriptions

observation

- Qualitative
  - color: black
  - animal: dog
- Quantitative
  - Discrete
    - tail: 1
    - legs: 4
  - Continuous
    - weight: 2.35 kg
    - height: 19.45 cm

MEASUREMENTS

Level of Measurements

Stevens's Topology, 1946 by Stanley Smith Stevens

nominal ordinal interval ratio

nominal

Used for labeling variables

- parts of speech: noun, verb, adjective etc.
- sector: bank, finance, insurance etc.

ordinal

ordered categories, inbetween distance (size) unknown

- speed: crawl, slow, average, fast, tooo much
- age: baby, teenager, youth, middle, old

interval

categories in numeric scales

- temperature: -100^{o}C, 0^{o}C, 100^{o}C
- time: 09:00, 16:00, 20:30

ratio

relation of to interval category.

```
We don't compare 8000 ko stock and 500 ko stock directly, we use ratios. Eg using log to
change the scale
```

summary

	nominal	ordinal	interval	ratio
property	class	compare	affinity	magnitu
direction	no	yes	yes	yes
size	no	no	yes	yes
zero	no	valid	vaild	invalid

HOW TO SHOW

statistically

- data summary
  - center of variation
  - summary of variation
  - range of variation

visualization

- plotting

  Many a time, we need to visually represent the data for both exploration and reporting.

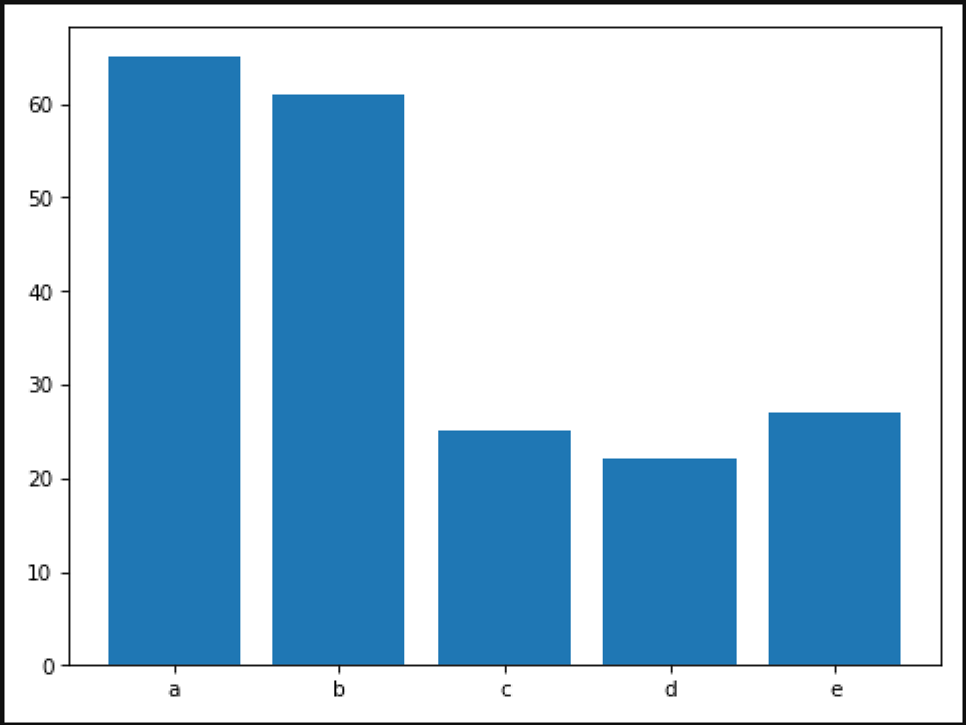
BAR

where

- show relative sizes
- show in categories

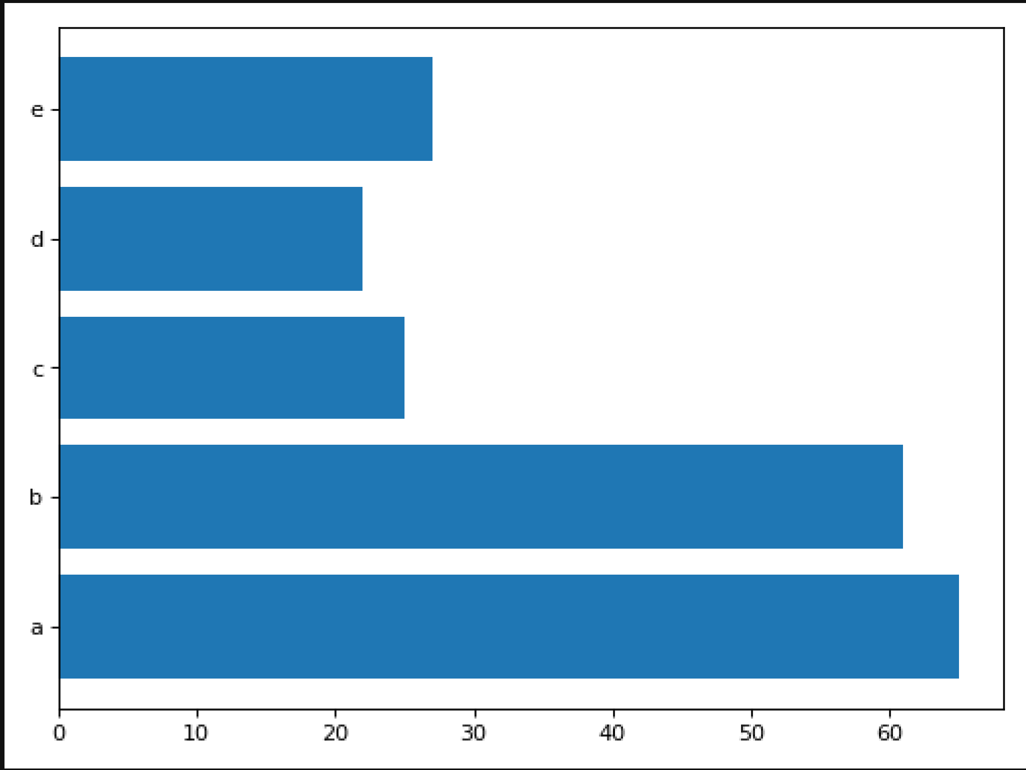
ax.bar()

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots(layout='tight')
ax.bar(list('abcde'), [65, 61, 25, 22, 27])
plt.show()
```



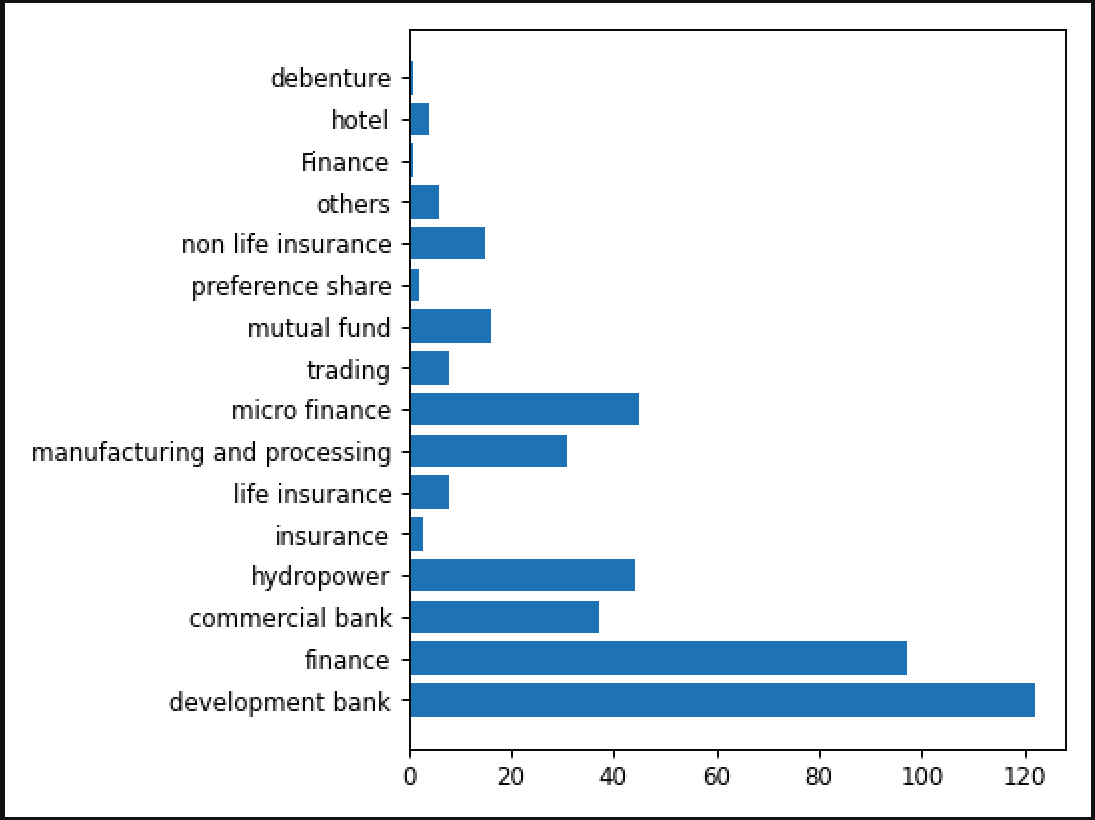
ax.barh()

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots(layout='tight')
ax.barh(list('abcde'), [65, 61, 25, 22, 27])
plt.show()
```



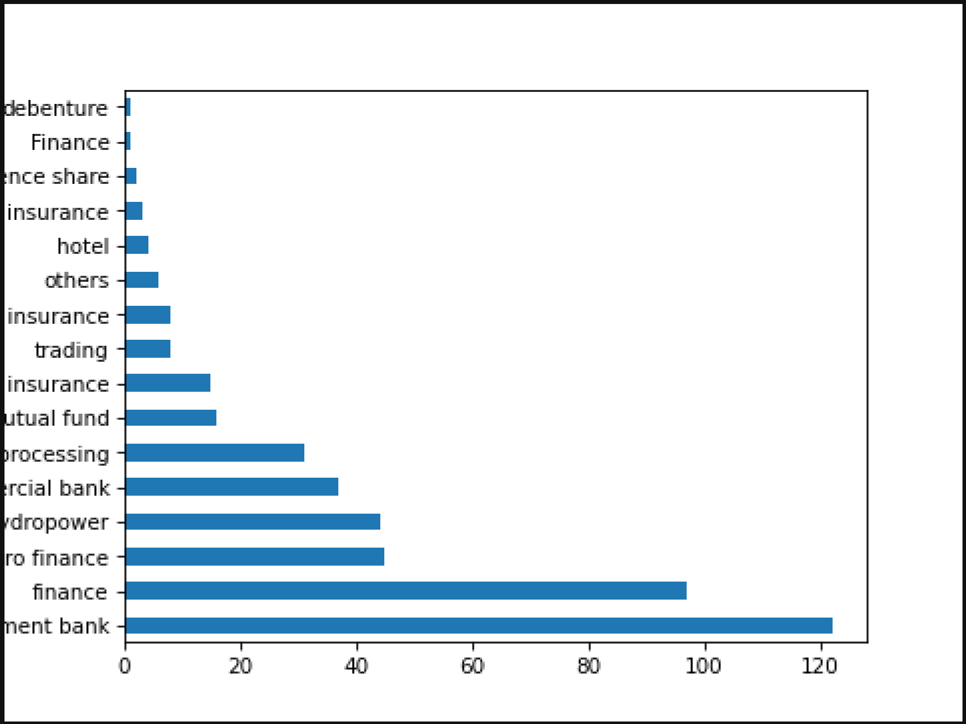
example

```
with open('./data/companies.tsv') as fp:
    for i, line in enumerate(fp):
        if i == 0: continue
        print(line.strip().split('\t'))
```

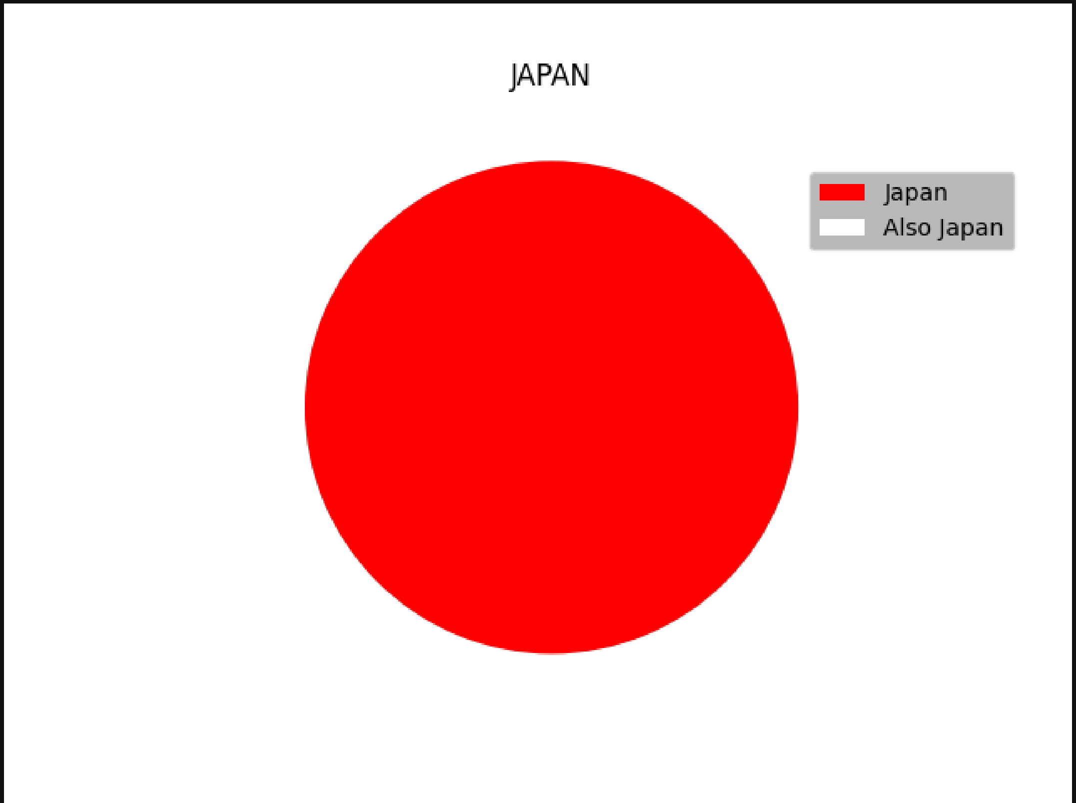


df.plot(kind='bar')

```
import pandas as pd
import matplotlib.pyplot as plt

df = pd.read_csv('./data/companies.tsv', delimiter='\t')
ax = df['sector'].value_counts().plot(kind='barh')
plt.show()
```



PIE

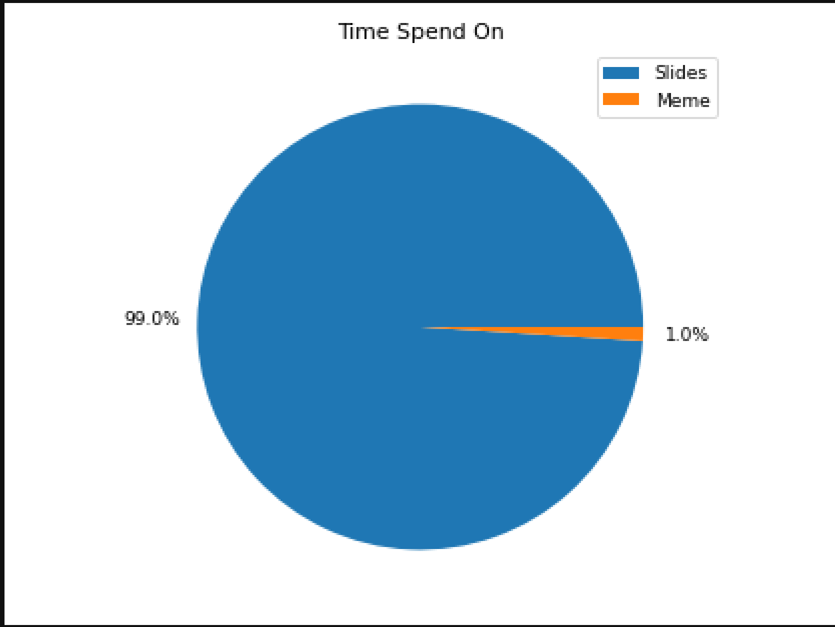


used for

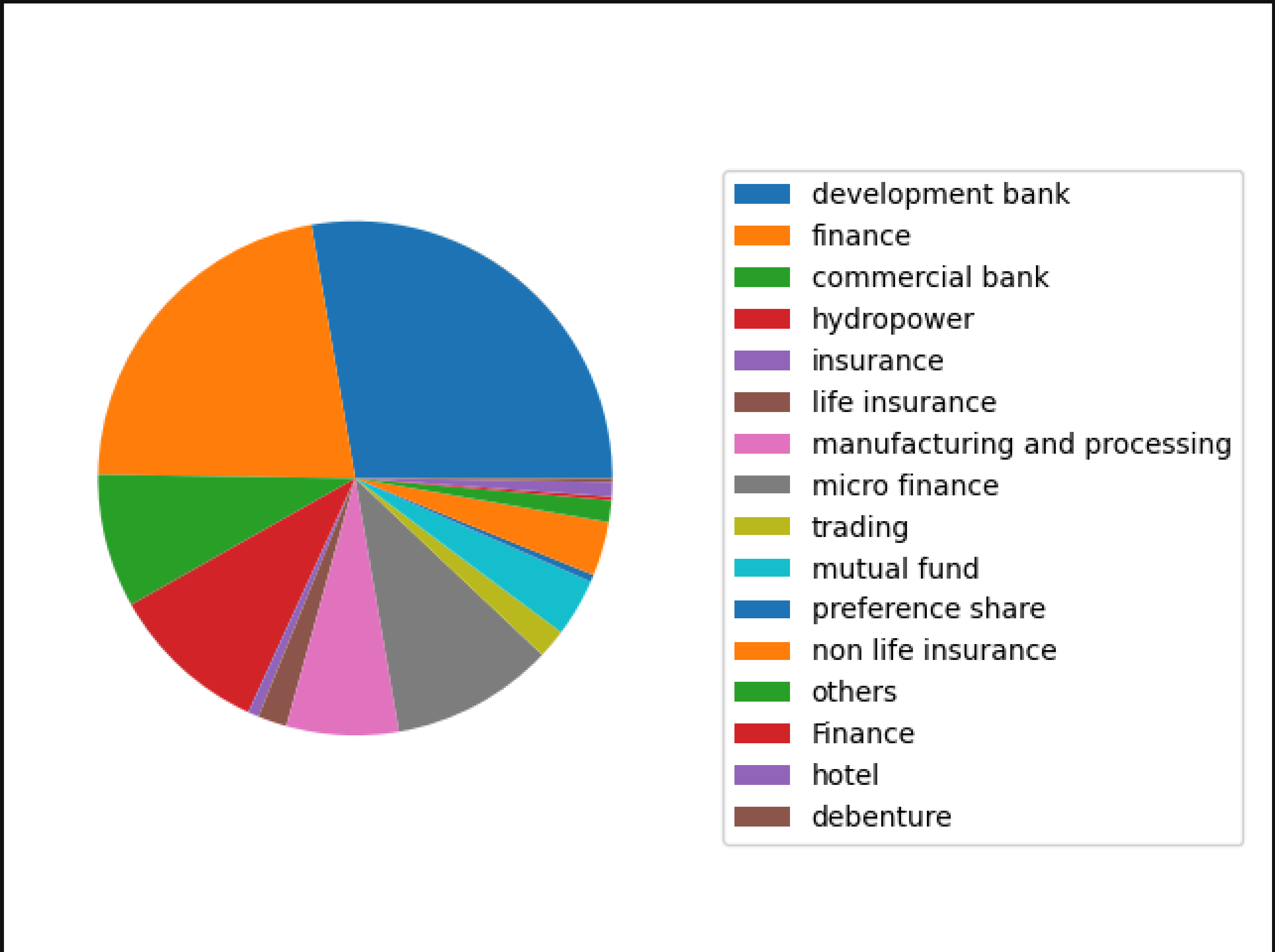
- relative sizes in proportion
- show in categories

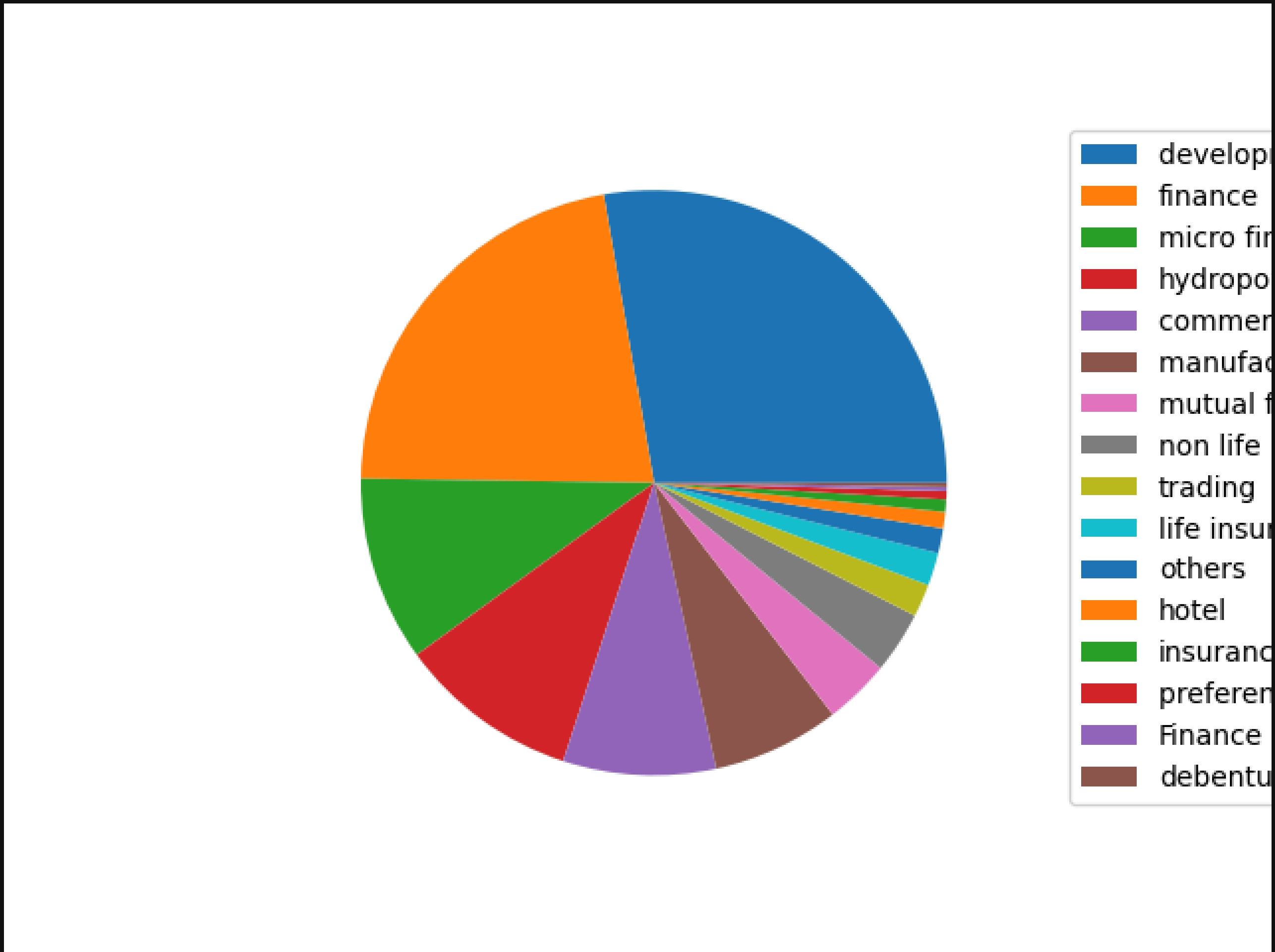
syntax

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots(layout='tight')
ax.pie([99, 1], autopct='%1.1f%%', pctdistance=1.2)
ax.set_title('Time Spend On')
plt.legend(labels=['Slides', 'Meme'], bbox_to_anchor=(0.8, 1))
plt.show()
```



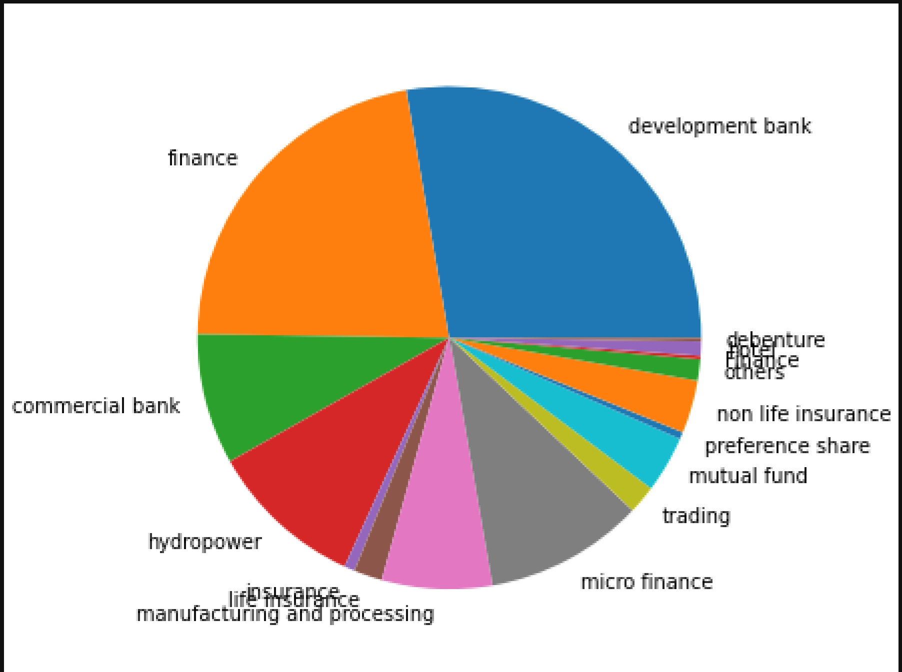
example


development bank
finance
commercial bank
hydropower
insurance
life insurance
manufacturing and processing
micro finance
trading
mutual fund
preference share
non life insurance
others
Finance
hotel
debenture

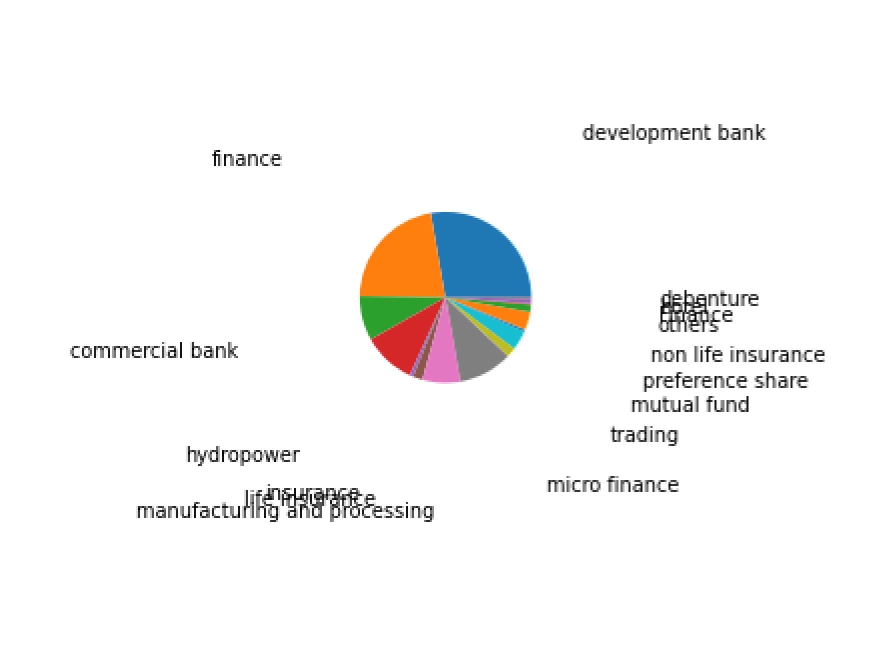

develop
finance
micro fir
hydropo
commer
manufac
mutual f
non life
trading
life insur
others
hotel
insuranc
preferen
Finance
debentu

ax.pie(..., labels)

```python
import matplotlib.pyplot as plt

fig, ax = plt.subplots(layout='tight')
ax.pie(freq.values(), labels=freq.keys())
plt.show()
```

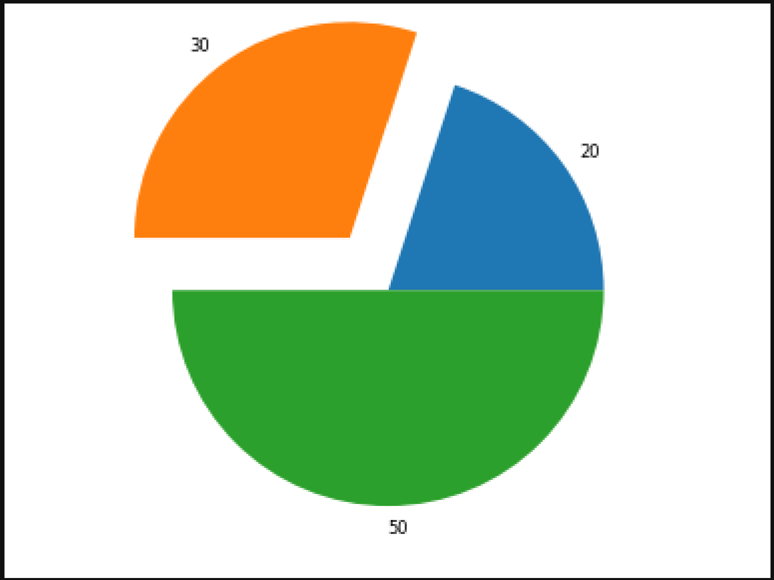


```python
import matplotlib.pyplot as plt

fig, ax = plt.subplots(layout='tight')
ax.pie(freq.values(), labels=freq.keys(), labeldistance=2.5)
plt.show()
```



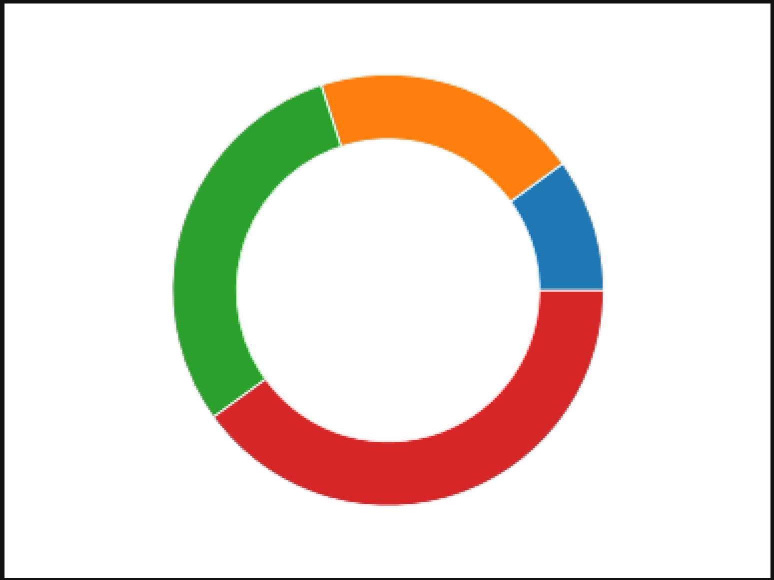
explode

```
import matplotlib.pyplot as plt

x = [20, 30, 50]
fig, ax = plt.subplots(layout='tight')
ax.pie(x, labels=x, explode=[0, .3, 0])
plt.show()
```



doughnut

```
import matplotlib.pyplot as plt

fig, ax = plt.subplots(layout='tight')
ax.pie(range(1, 5), wedgeprops={
    'width' : 0.3,
    'edgecolor' : 'white'
})
plt.show()
```



CENTER OF VARIATION

types

also know as averages

- mean
- median
- mode

average

$$\mu = \frac{1}{N} \sum_{i=0}^{N} x_i$$

```python
print(sample := list(range(10)))
print(sum(sample)/len(sample))

import statistics  # new in version 3.4
print(statistics.mean(range(10)))

import numpy as np
print(np.mean(sample))
```

```
[0, 1, 2, 3, 4, 5, 6, 7, 8, 9]
4.5
4.5
4.5
```

in the middle

$$\mathrm{median}(x) = \frac{1}{2}\left(x_{\lfloor (n+1)/2 \rfloor} + x_{\lceil (n+1)/2 \rceil}\right)$$

```python
import statistics
import numpy as np

print(a:=[1, 2, 3, 4, 9], statistics.median(a))
print(b:=[2, 1, 8, 9], np.median(b))
print(c:=[9, 7, 8, 1], np.median(c))
```

```
[1, 2, 3, 4, 9] 3
[2, 1, 8, 9] 5.0
[9, 7, 8, 1] 7.5
```

most repeated

```python
import statistics

print(a:=range(10), statistics.mode(a))  # no mode
print(b:=[1, 3, 3, 4, 9], statistics.mode(b))
print(c:=[9, 3, 3, 9, 4], statistics.mode(c))
```

```
range(0, 10) 0
[1, 3, 3, 4, 9] 3
[9, 3, 3, 9, 4] 9
```

effect of outliers

sensitivity of data

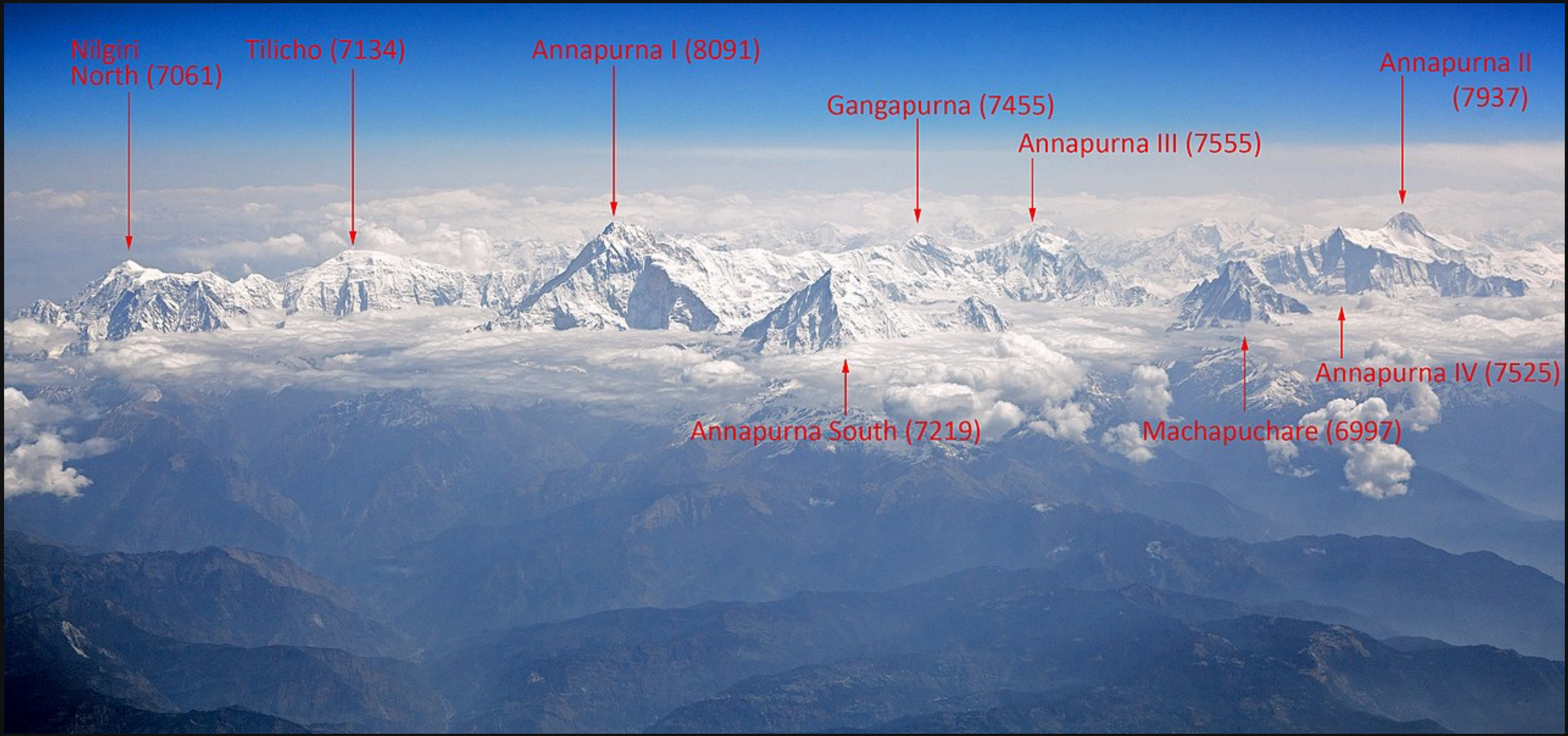
```python
import statistics

sample = [ 1, 1, 99999, 1 ]
print(statistics.mean(sample))
print(statistics.median(sample))
print(statistics.mode(sample))
```

```
25000.5
1.0
1
```

summary

	nominal	ordinal	interval	ratio
mean	no	no	yes	yes
median	no	yes	yes	yes
mode	yes	yes	yes	yes

RANGE OF VARIATION



types

understanding the spread of the data

- range
- mid-range

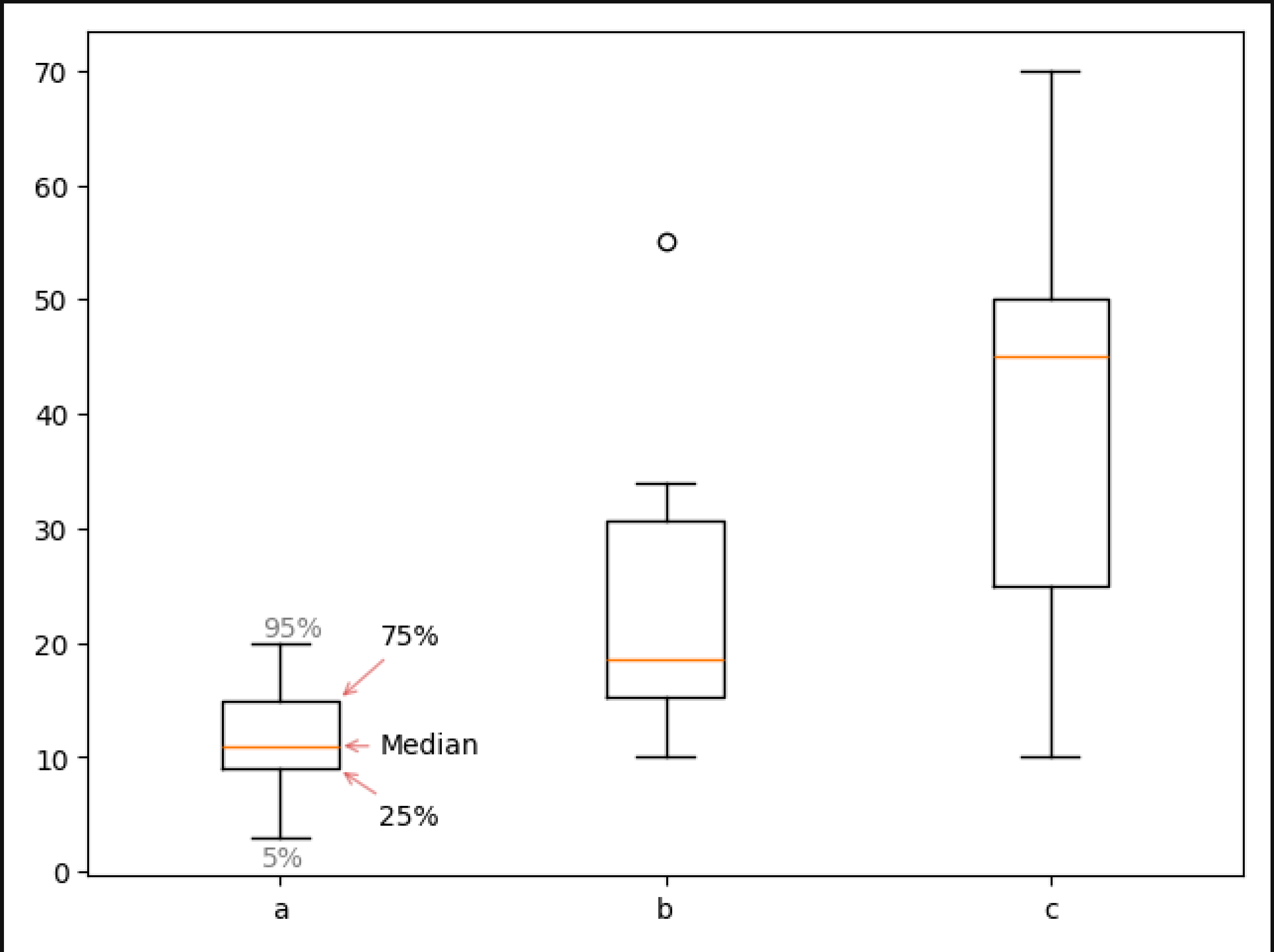
range

$$\mathrm{range}(x) = \max(x) - \min(x)$$

```python
import numpy as np

a = np.array([2, 5, 3, 4, 5, 3])
print(a.max() - a.min())
print(np.ptp(a))
```

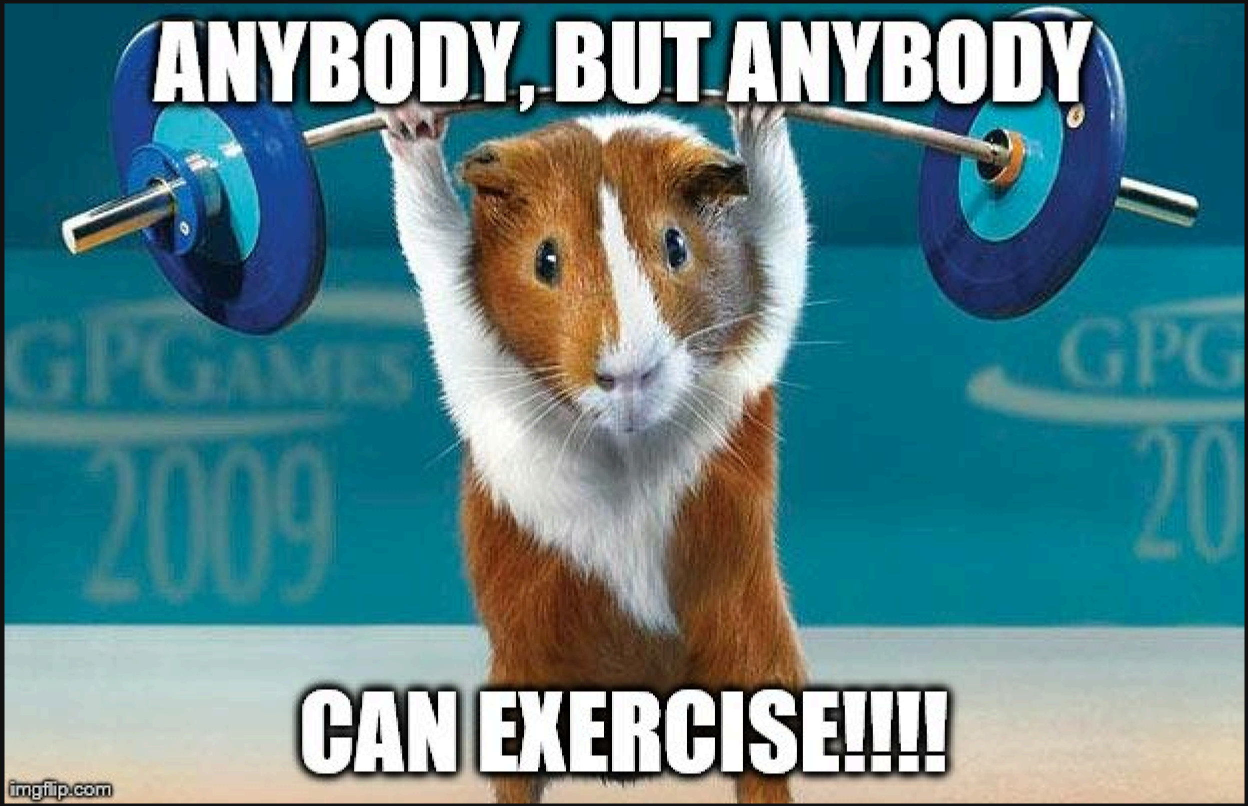
```
3
3
```

box and whisker



```python
import matplotlib.pyplot as plt

data = {
    'a' : [ 3, 5, 5, 9, 10, 11, 11, 12, 12, 15, 18, 18, 20 ],
    'b' : [ 10, 15, 16, 21, 34, 55 ],
    'c' : [ 10, 25, 25, 44, 45, 49, 50, 55, 70 ],
}

fig, ax = plt.subplots(layout='tight')
bp = ax.boxplot(
    data.values(),
    tick_labels = data.keys(),
)
```

EXERCISE



data

time in min to get to school on 12 separate days

bus	walking
16	22
14	19
15	21
14	20
31	21
15	20

load

```python
import numpy as np

d = np.array(data)
print(d)
```

```
[[16 22]
 [14 19]
 [15 21]
 [14 20]
 [31 21]
 [15 20]]
```

question 1

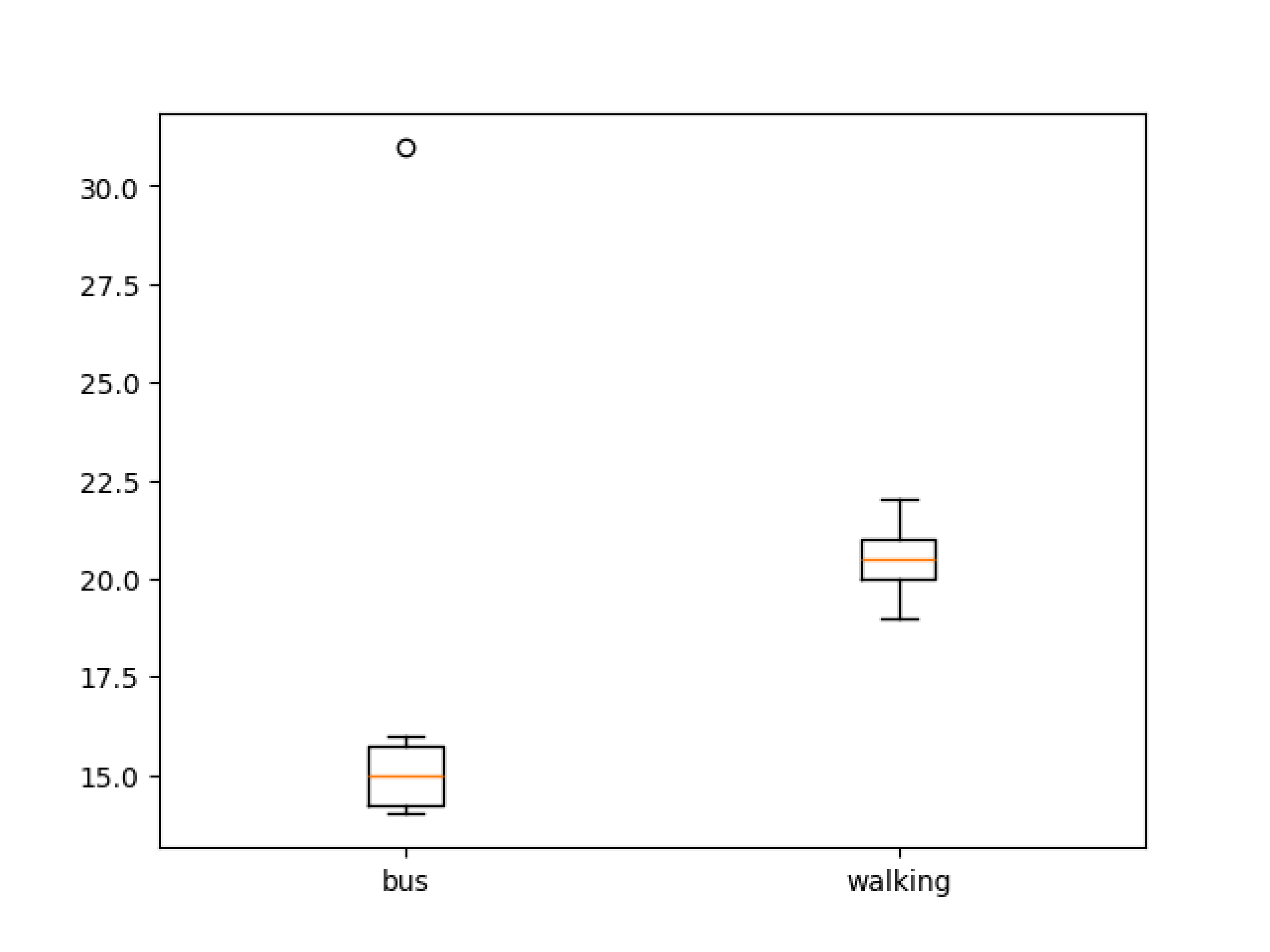
Which mode of travel is faster?

question 2

if school is starting in next 25m, how should we travel so we won't be late.

question 3

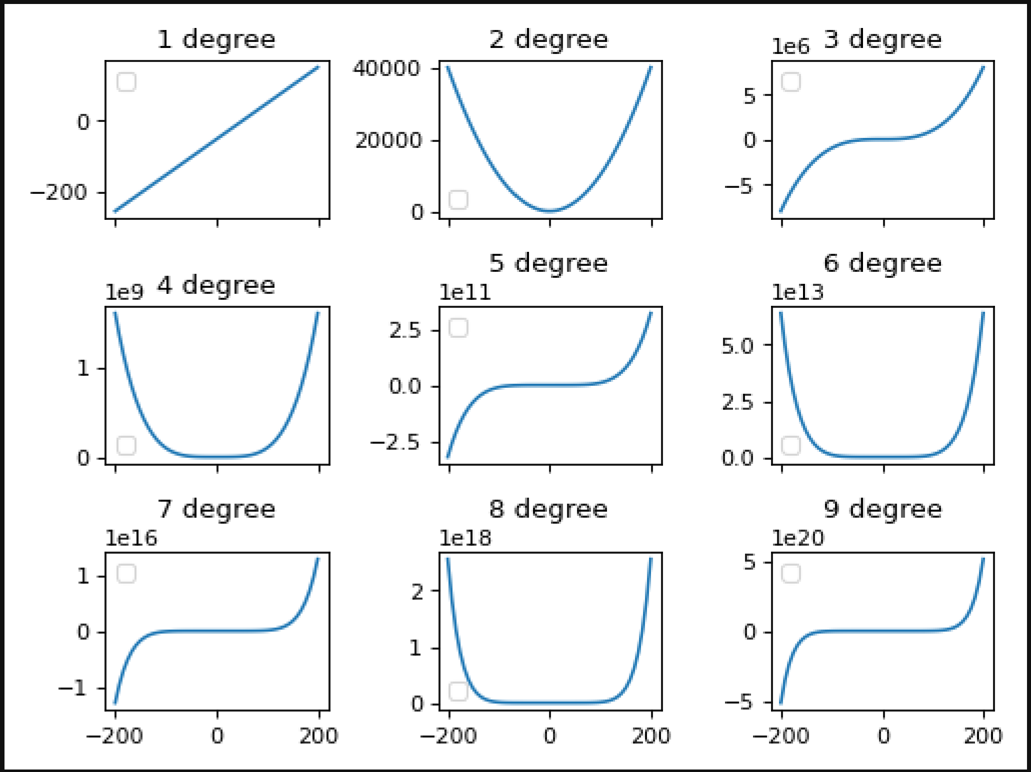
box plot the data



LINE

plot

$$y = x^2 + 1$$

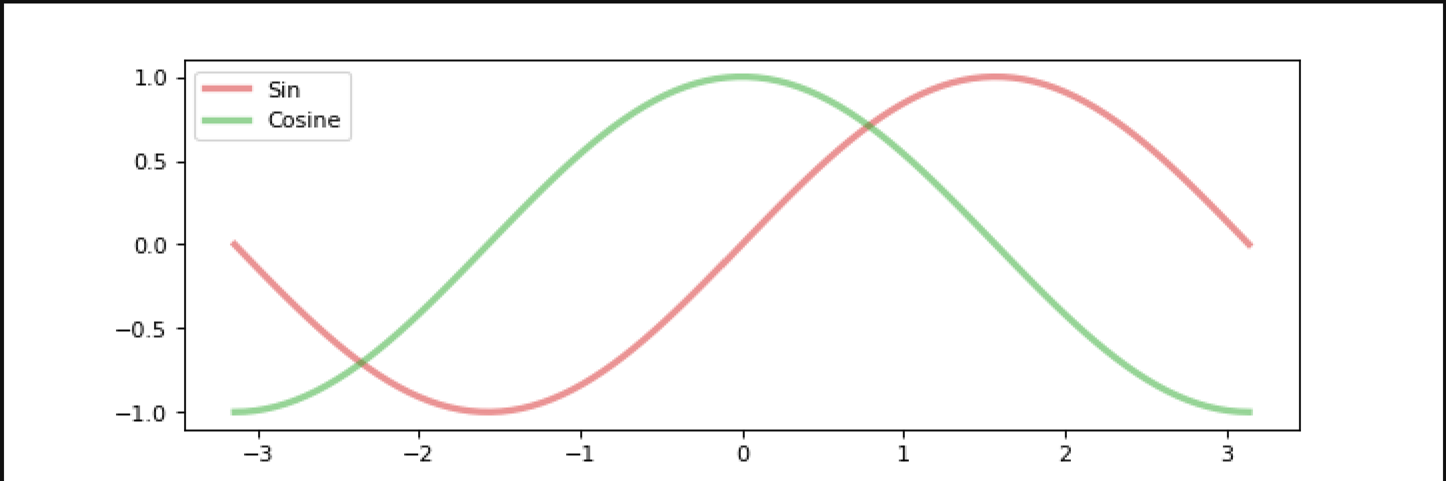


1 degree
2 degree
1e6 3 degree
1e9 4 degree
5 degree
1e11
6 degree
1e13
7 degree
1e16
8 degree
1e18
9 degree
1e20
−200
0
200

waves

```python
import numpy as np
print(np.sin(np.radians(90)))
print(np.linspace(1, 2, 5))
```

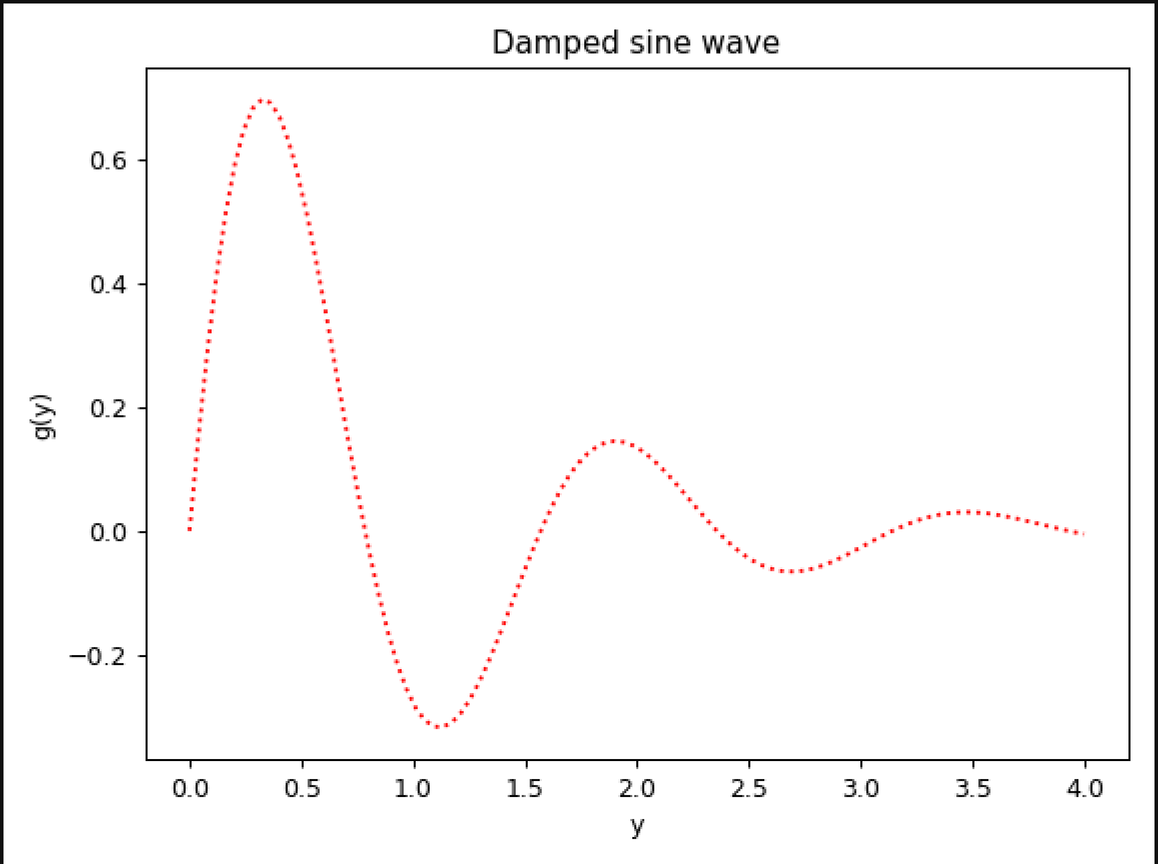
```
1.0
[1.   1.25 1.5  1.75 2.  ]
```



```python
import numpy as np
import matplotlib.pyplot as plt

fig, ax = plt.subplots(figsize=(9, 3))

x = np.linspace(-np.pi, np.pi, 100, endpoint=True)
ax.plot(x, np.sin(x), color='#d627287f', lw=3, label='Sin')
ax.plot(x, np.cos(x), color='#2ca92c7f', lw=3, label='Cosine')
plt.legend()
plt.show()
```

damping

$$g(y) = e^{-y}\sin(4y) \text{ for } y \in [0, 4]$$



```python
import numpy as np
import matplotlib.pyplot as plt

y = np.linspace(0, 4, 100)
g = lambda y: np.exp(-y) * np.sin(4*y)

fig, ax = plt.subplots(layout='tight')
plt.plot(y, g(y), color='red', linestyle='dotted')
plt.xlabel('y')
plt.ylabel('g(y)')
plt.title('Damped sine wave')
plt.show()
```

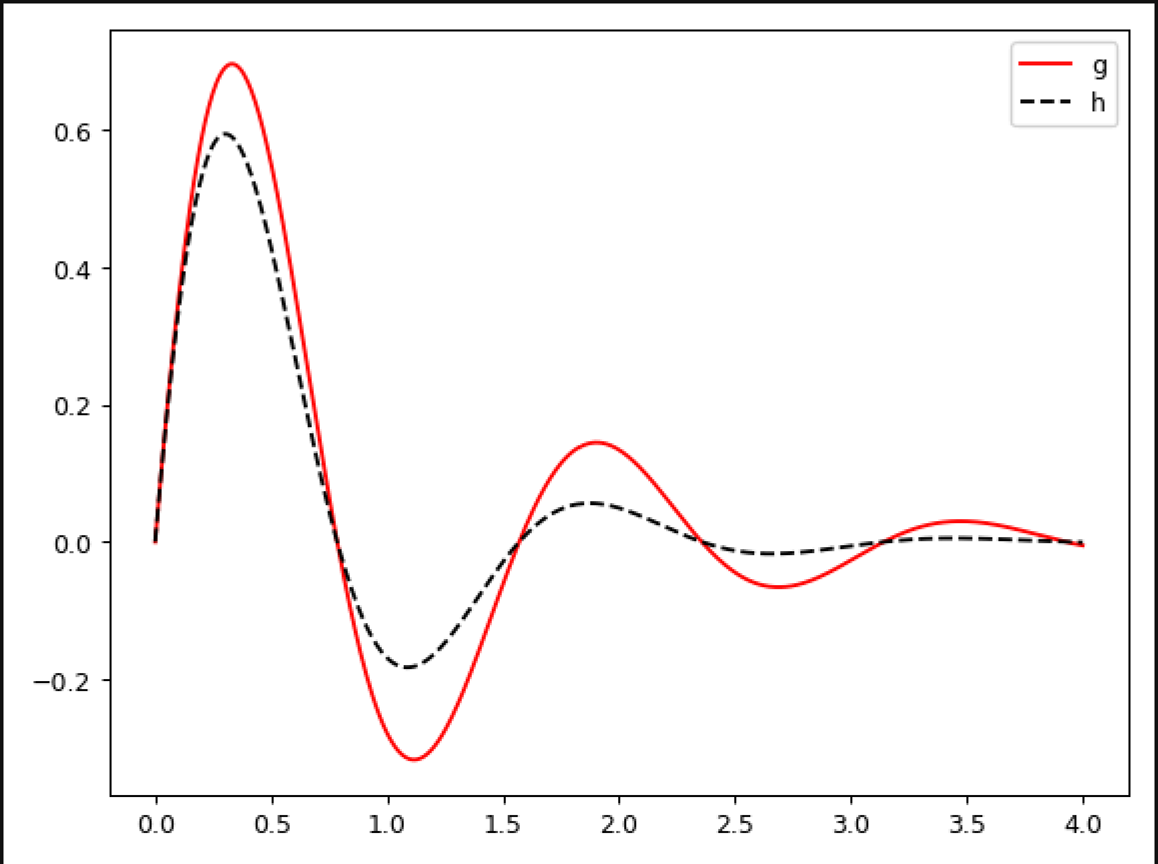
linestyle

default format is 'b-'

long	short
solid	'-'
dashed	'–'
dashdot	'-.'
dotted	':'

multiple

$g(y) = e^{-y}\sin(4y)$ and $h(y) = e^{-\frac{3}{2}y}\sin(4y)$ for $y \in [0, 4]$


g
h
0.6
0.4
0.2
0.0
−0.2
0.0
0.5
1.0
1.5
2.0
2.5
3.0
3.5
4.0

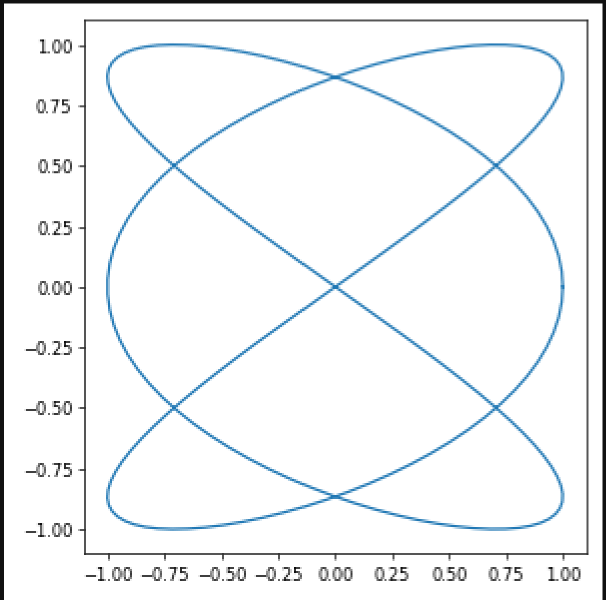
```python
import numpy as np
import matplotlib.pyplot as plt

y = np.linspace(0, 4, 501)

fig, ax = plt.subplots(layout='tight')
ax.plot(y, np.exp(-y) * np.sin(4*y), 'r-')
ax.plot(y, np.exp(-(3/2)*y) * np.sin(4*y), 'k--')
ax.legend(['g', 'h'])
plt.show()
```

Lissajous Curve

$$x = \sin(at + \delta) \qquad y = \sin(bt)$$

for values $a = 3, \quad b = 2, \quad \delta = \pi/2$

between $t \in [0, 2\pi]$, with 100 sampling for t


1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
−0.75
−1.00
−1.00 −0.75 −0.50 −0.25 0.00 0.25 0.50 0.75 1.00

```python
import numpy as np
import matplotlib.pyplot as plt

a, b, delta = 3, 2, np.pi/2
t = np.linspace(0, 2*np.pi, 200)

fig, ax = plt.subplots(layout='tight', figsize=(5, 5))
ax.plot(np.sin(a*t + delta), np.sin(b*t))
plt.show()
```

SUMMARY OF VARIATION

types

- standard deviation
- quantiles/percentile

visualization: box-plot

standard deviation

$$\sigma^2 = \frac{1}{N - \mathrm{ddof}} \sum_{i=0}^{N} (x_i - \mu)^2 \qquad s = \sqrt{\sigma}$$

```python
import statistics

l = list(range(10))
print(statistics.variance(l))
print(statistics.stdev(l))
print(statistics.pvariance(l))
print(statistics.pstdev(l))
```

```
9.166666666666666
3.0276503540974917
8.25
2.8722813232690143
```

```python
import numpy as np

l = list(range(10))
print(np.var(l, ddof=1))
print(np.std(l, ddof=1))
print(np.var(l))
print(np.std(l))
```

```
9.166666666666666
3.0276503540974917
8.25
2.8722813232690143
```

quantiles/percentile

is a value of threshold for a variable

```
import statistics
```

```
import numpy as np

print(np.percentile(range(101), 90))
print(np.percentile(range(11), 10))
```

```
90.0
1.0
```